COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 2

PREDICATES AND QUANTIFIERS

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PREDICATES

- ***Predicate***: A property that the subject of the statement can have. A predicate is a sentence that contains a finite number of variables and becomes a proposition when specific values are substituted for the variables.

  Example: " x > 3 " here x: variable and >3: predicate

- A PREDICATE is symbolized by a CAPITAL LETTER and the variable(s) by small letter(s).
- We can denote the statement "x is greater than 3" by $P(x)$, where P denotes the predicate "is greater than" and x is the variable. The statement $P(x)$ is also said to be the value of the propositional function P at x.

  $P(x): x>3$

- A ***predicate*, or propositional function**, is a function that takes some variable(s) as arguments and returns True or False.
- **Note**: *Once a value has been assigned to the variable x, the statement P(x) becomes a proposition and has a truth value (either TRUE or FALSE)*

PREDICATES

Q. Let *P*(x) denote the statement "x>3". What are the truth values of *P*(4) and *P*(2)?

Solution: Given ==> *P*(x) :"x>3"

- We obtain the statement *P*(4) by setting x=4 in the statement "x>3". Hence *P*(4), which is the statement "**4>3**", is **true**.
- However, *P*(2) which is the statement "**2>3**", is **false**.

Q. Let, *A*(x) :"Computer x is under attack by an intruder". Suppose that of the computers on campus, only C1 and C7 are currently under attack by intruders. What are the truth values of *A* (C1), *A*(C3), *A*(C7)?

Solution:

- *A*(C1):"Computer C1 is under attack by an intruder" is true
- *A*(C7): "Computer C7 is under attack by an intruder" is true
- *A*(C3):"Computer C3 is under attack by an intruder" is false (C3 is not in the list)

MULTIVARIABLE PREDICATES

- *Multivariable* Predicates ==> Predicates that have more than one variable.
- For example, $Q(x, y)$: "$x = y + 3$" ,

  where x and y are variables and Q is the predicate.
- Note: When values are assigned to the variables x and y, the statement $Q(x, y)$ has a truth value.

Q. Let $Q(x, y)$ denote the statement "$x = y + 3$". What are the truth values of the propositions $Q(1,2)$ and $Q(3,0)$?

Solution:

To obtain $Q(1,2)$, set x=1 and y=2 in the statement Q(x, y).

Therefore, $Q(1,2)$: "1=2+3" is **false**

Similarly, $Q(3,0)$: "3=0+3" is **true**

QUANTIFIERS

- **Quantification:**
  - Universal quantification: A predicate is true for every element

    "∀" reads "for **A**ll"
  - Existential quantification: There is one or more elements for which a predicate is true

    "∃" reads "there **E**xists"
- Domain/domain of discourse/universe of discourse: The values a variable in a *propositional function* may take.
- Once a variable has been bound, we cannot bind it again. For example the expression

  $\forall x\ (\ \forall x\ P\ (x)\ )$ is nonsensical.
- The interior expression ($\forall x\ P\ (x)$) bounded x already and therefore made it unobservable to the outside.

UNIVERSAL QUANTIFIERS

- **Definition**: The *universal quantification* of *P(x)* is the statement "*P(x)* for all values of x in the domain".
- The notation $\forall x\, P(x)$ denotes the universal quantification of *P(x)*.
- $\forall$ is called the universal quantifier
- We read $\forall x\, P(x)$ **as** "for all *x*, *P(x)*" or "for every *x*, *P(x)*"
- An element for which *P*(x) is false is called a **counterexample** of $\forall x\, P(x)$
- "$\forall x\, P\,(x)$" true when ***every instance*** of *x* makes *P* (*x*) true when plugged in
- Like taking **conjunction** over entire universe:

$$\forall x\, P\,(x\,) \equiv P\,(x_1) \wedge P\,(x_2) \wedge P\,(x_3) \wedge \ldots \wedge P(x_n)$$

UNIVERSAL QUANTIFIERS

Q: What is the truth value of $\forall x\, P(x)$, where P(x) is the statement "$x^2<10$" and the domain consists of the positive integers less than 4?

Solution: The statement $\forall x\, P(x)$ is the same as the conjunction **$P(1) \wedge P(2) \wedge P(3)$,** because the domain consists of the integers 1, 2, 3. So, **$\forall x\, P(x)$ is true.**

How?----------See Below----------

P(1): "$1^2<10$", is true

P(2): "$2^2<10$" is true

P(3): "$3^2<10$" is true

$P(1) \wedge P(2) \wedge P(3) \rightarrow T \wedge T \wedge T \rightarrow T$

UNIVERSAL QUANTIFIERS

Q: Let $P(x)$ be the statement "$x+1>x$" What is the truth value of the quantification $\forall x\, P(x)$, where the domain consists of all real numbers?

Solution: Because $P(x)$ is true for all real numbers x, the quantification $\forall x\, P(x)$ is **true**.

Q: Let $Q(x)$ be the statement "$x<2$". What is the truth value of the quantification $\forall x\, Q(x)$, where the domain consists of all real numbers?

Solution: $Q(x)$ is not true for every real number x, because, for instance, $Q(3)$ is false. That is, $x=3$ is a counterexample for the statement $\forall x\, Q(x)$. Thus, $\forall x\, Q(x)$, is false

Q: Let P(x) be the statement "$x^2>0$". What is the truth value of the quantification $\forall x\, P(x)$, where the universe of discourse consists of all integers?

Solution: P(x) is not true for all integers. We can give a counter example. We see that $x=0$ is a counterexample, because $x^2 = 0$ when $x = 0$, so that x^2 is not greater than 0 when $x = 0$ Therefore, $\forall x\, P(x)$ **is false**.

EXISTENTIAL QUANTIFIERS

- **Definition**: The existential quantification of *P(x)* is the proposition "There exists an element *x* in the domain such that *P(x)*".
- We denote the existential quantification of *P(x)* by $\exists x\, P(x)$
- $\exists$ is called the **existential quantifier**.
- Existential quantification $\exists x\, P(x)$ is read as:
  - "There is an *x* such that *P(x)*",
  - "There is at least one x such that P(x)", or
  - "for some *x* *P(x)*"
- "$\exists x\, P(x)$" is true when *an instance* can be found which when plugged in for *x*, makes *P (x)* true.
- Like taking **disjunction** over entire universe

$$\exists x\, P(x) \equiv P(x_1) \vee P(x_2) \vee P(x_3) \vee \ldots P(x_n)$$

EXISTENTIAL QUANTIFIERS

Q: Let $P(x)$ denote the statement "x>3". What is the truth value of the quantification $\exists x\, P(x)$", where the domain consists of all real numbers?

Solution: Because "x>3" is sometimes true for instance, when x=4, the existential quantification of $P(x)$, which is $\exists x\ P(x)$, **is true**.

Q: Let $Q(x)$ denote the statement "x=x+1". What is the truth value of the quantification $\exists x\, Q(x)$, where the domain consists of all real numbers?

Solution: Because $Q(x)$ is false for every real number x, the existential quantification of $Q(x)$, which is $\exists x\, Q(x)$, is **false**.

Q: What is the truth value of $\exists x\, P(x)$, where P(x) is the statement "$x^2>10$" and the universe of discourse consists of the positive integers not exceeding 4?

Solution: Because the domain is { 1, 2, 3, 4}, the proposition $\exists x\, P(x)$ is the same as the **disjunction** $P(1) \vee P(2) \vee P(3) \vee P(4)$.

Because $P(4)$, which is the statement "$4^2>10$" , is true, it follows that **$\exists x\, P(x)$ is true**.

EXISTENTIAL QUANTIFIERS

Q: Let *P*(x) denote the statement "x>0". What is the truth value of the quantification $\exists x\, P(x)$", where the domain consists of integers? True

Q: Let *P*(x) denote the statement "x>0". What is the truth value of the quantification $\forall x\, P(x)$", where the domain consists of non-negative integers? False

Q: Let *P*(x) denote the statement "x>0". What is the truth value of the quantification $\exists x\, P(x)$", where the domain consists of negative integers? False

Q: Let *P*(x) denote the statement "x<2". What is the truth value of the quantification $\exists x\, P(x)$", where the domain consists of all prime numbers? False

Q: Let *P*(x) denote the statement "$x \leq 2$". What is the truth value of the quantification $\exists x\, P(x)$", where the domain consists of all prime numbers? True

TRANSLATING *FROM* ENGLISH *INTO* LOGICAL EXPRESSIONS

Q: Express the statement "Every student in the class has studied calculus" using predicates and quantifiers.

Solution: First, we rewrite the statement so that we can clearly identify the appropriate quantifiers to use. Doing so, we obtain:

"For every student in the class, that student has studied calculus".

Next we introduce a variable x so that our statement becomes –

"For every student x in the class, x has studied calculus"

Continuing, we introduce the predicate $C(x)$, which is the statement "x has studied calculus"

Consequently, if the universe of discourse for x consists of the students in the class, we can translate our statement as $\forall x\, C(x)$.

TRANSLATING *FROM* ENGLISH *INTO* LOGICAL EXPRESSIONS

Note: There are other correct approaches; different domains of discourse and other predicates can be used. For example,

If we **change** the **domain** to consists of **all people**, we need to express our statement as

"For every person x, if person x is a student in this class then x has studied calculus."

If S(x) represents the statement that person x is in this class, our statement can be expressed as

$\forall x\, S(x) \rightarrow C(x)$.

***Note**: For the second way, we always want to use **conditional statements** with **universal quantifiers** and **conjunctions** with **existential quantifiers**.*

TRANSLATING *FROM* ENGLISH *INTO* LOGICAL EXPRESSIONS

Q: Express the statement "Someone in your school has studied calculus" using predicates and quantifiers. Let the **domain** (universe of discourse) consists of **all people.**

Solution:

Let, S(x) be the propositional functions " x is in your school" and

C(x) be the propositional function "x has studied calculus".

$$\exists x\ (\ S(x) \wedge C(x)\ )$$

- **Note that** if the domain consists of the students in your school, then we can write $\exists x\ C(x)$

TRANSLATING *FROM* ENGLISH *INTO* LOGICAL EXPRESSIONS

Q: Let *P(x)* be the statement "*x* can speak Russian" and let *Q(x)* be the statement "*x* knows the computer language C++", Express each of these sentences in terms of *P(x)*, *Q (x)*, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

- There is a student at your school who can speak Russian and who knows C++.

  $\exists x (p(x) \wedge Q(x))$

- There is a student at your school who can speak Russian but who doesn't know C++.

  $\exists x (p(x) \wedge \neg Q(x)]$

- Every student at your school either can speak Russian or knows C++.

  $\forall x (P(x) \vee Q(x))$

- No student at your school can speak Russian or knows C++

  $\forall x \neg (P(x) \vee Q(x))$

TRANSLATING *FROM* ENGLISH *INTO* LOGICAL EXPRESSIONS

Q: Translate each of the statements into logical expressions using predicates, quantifiers, and logical connectives.

- No one is perfect. $\forall x \neg P(x)$
- Not everyone is perfect. $\neg \forall x P(x)$
- All your friends are perfect. $\forall x (F(x) \rightarrow P(x))$
- At least one of your friends is perfect. $\exists x (F(x) \wedge P(x))$
- Everyone is your friend and is perfect. $\forall x (F(x) \wedge P(x))$ *or* $(\forall x (F(x)) \wedge (\forall x P(x))$
- Not everybody is your friend or someone is not perfect. $(\neg \forall x (F(x)) \vee (\exists x \neg P(x))$

REFERENCES

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- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley